Aatman Pandya

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Professional Summary

Data Science and AI enthusiast with experience in **Machine Learning**, **NLP**, analytics, and **Python**-based application development. Skilled in building predictive models, semantic search systems, and data-driven decision support tools using **Python**, **SQL**, machine learning, and modern NLP techniques. Interested in applying AI and analytics to solve real-world business problems through intelligent automation and data-driven insights.

Education

BITS Pilani, Hyderabad Campus — Hyderabad, India
MSc. (Hons) Biological Science & B.E. (Hons) Mechanical Engineering — *Expected May 2027*

Experience

Data Analytics Intern — May 2024 – Jul 2024
Octalsoft

- Developed **Python** analytics workflows using Pandas and NumPy to process and analyze 1,000+ customer records, account balances, completion status, and operational performance indicators.
- Built automated reporting pipelines using data cleaning, validation, and exploratory analysis techniques to support business monitoring, anomaly detection, and operational decision-making.
- Generated analytical reports and dashboard-ready insights to track completion trends, balance irregularities, and key operational metrics across customer workflows.
- Developed **Oracle APEX** front-end modules for enterprise management systems, supporting data-entry, reporting, and operational tracking functionality.

Projects

AI Resume ATS Analyzer | *Python, NLP, Sentence Transformers, Embeddings, Semantic Search* — Jan 2026 – Mar 2026

- Built an AI-powered Applicant Tracking System using modular document ingestion, PDF parsing, resume extraction, embedding generation, semantic retrieval, scoring, and recommendation workflows.
- Implemented **Sentence Transformers** using all-MiniLM-L6-v2 embeddings and cosine similarity scoring to enable semantic resume-job matching, contextual retrieval, and candidate ranking beyond keyword overlap.
- Integrated LLM-based information extraction, named entity recognition, fuzzy matching, and structured parsing to convert unstructured resumes into comparable candidate profiles.
- Designed retrieval workflows and ATS recommendation logic using semantic search signals to surface skill gaps, improve resume relevance, and generate targeted optimization suggestions.

Smartwatch Price Prediction System | *Python, Scikit-Learn, XGBoost, Random Forest* — Oct 2025 – Dec 2025

- Developed an end-to-end supervised ML pipeline for smartwatch price prediction using 450 smartwatch records, incorporating feature engineering, preprocessing, model training, benchmarking, and evaluation.
- Applied missing-value imputation, IQR-based outlier removal, one-hot encoding, and Min-Max scaling to transform raw smartwatch attributes into a 201-feature training dataset for predictive modeling.
- Benchmarked Decision Tree, **Random Forest**, and **XGBoost** regressors, comparing predictive performance across multiple evaluation metrics to identify robust pricing models.
- Optimized model parameters with **GridSearchCV** and **K-Fold Cross Validation**, using R-squared metrics, feature importance analysis, and model interpretability checks to assess generalization performance.

Market Sentiment Analysis using FinBERT | *Python, FinBERT, NLP, Sentiment Analysis, Statistical Analysis* — Aug 2025 – Oct 2025

- Built a financial sentiment analysis pipeline using **FinBERT** to analyze 1,000+ financial news articles and quantify market sentiment across multiple publicly traded companies.
- Collected and processed approximately 1,400 news articles over a 30-day period, generating sentiment signals for five major technology stocks.
- Performed correlation analysis between sentiment scores and subsequent stock-price movements, identifying predictive relationships with correlation coefficients as high as 0.63.
- Developed visualization dashboards and statistical reporting workflows to transform unstructured financial news into decision-support insights for market analysis.

Technical Skills

Languages: **Python**, **SQL**, Java
Libraries: NumPy, Pandas, Scikit-Learn, Matplotlib
Machine Learning: **Regression**, **Classification**, Clustering, **Random Forest**, **XGBoost**, **Feature Engineering**, Hyperparameter Tuning, **Cross Validation**, **Model Evaluation**
AI/NLP: **Embeddings**, **Semantic Search**, **Sentence Transformers**, Cosine Similarity, Named Entity Recognition, NLP Pipelines
Tools: Git, GitHub, **Oracle APEX**
Coursework: Probability & Statistics, Data Analytics, Optimization

Leadership & Extracurriculars

Design Team Member — May 2023 – Jun 2025
Traders@BPHC — *BITS Pilani Hyderabad Campus*

- Created branding, marketing creatives, and promotional material to support club outreach, event visibility, and student engagement campaigns.

Logistics Coordinator — Jan 2025
Verba Maximus — *BITS Pilani Hyderabad Campus*

- Coordinated logistics, participant management, and event execution workflows to ensure smooth scheduling, venue operations, and on-ground coordination.